

COMPUTER SCIENCE AND ENGINEERING
(CYBER SECURITY)

CR23532

CYBER AND DIGITAL FORENSICS

ASSIGNMENT - 2

SUBMITTED BY

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ASSIGNMENT 2(1)

Suspected Unauthorized Access to a College Server

Case Background

A college administrator reports that confidential examination-related files may have been accessed and copied from a departmental server outside normal working hours. Potential evidence includes the complaint, a suspect workstation, server access logs, a USB device recovered from the workstation, and relevant files. The objective is to determine what happened while preserving the reliability of the evidence.

1. Identification, Collection, Examination, Analysis and Presentation

Identification. Investigators first confirm the allegation, define the scope, obtain written authorization, identify the systems and accounts involved, and record the reported time window. They identify likely sources: the server, suspect workstation, USB device, authentication logs, file-access logs, network/security logs, relevant examination files, and administrator statements. Each source is assigned an evidence identifier.

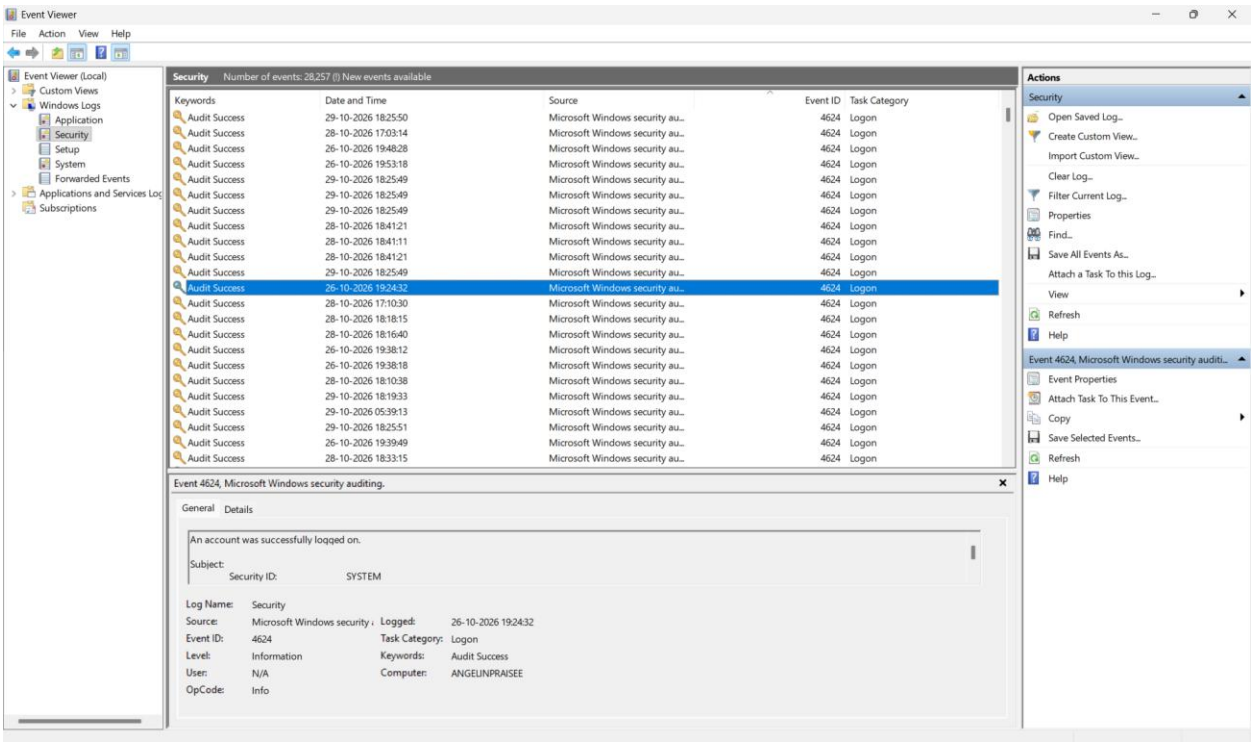
Collection. The team documents the physical condition, location, date/time, device make/model, serial number, user currently logged in, and network connections. Volatile data may be collected first when justified and authorized, because it can disappear when a device is shut down. Investigators then acquire forensic images of storage media using a validated process and write protection where applicable. Server logs are exported in a manner that preserves original records and metadata. Hash values are calculated for acquired images and relevant exported files.

Examination. Analysts work from forensic copies rather than original media. They inspect file-system metadata, event logs, authentication records, USB connection artifacts, recent-file records, browser/download artifacts if relevant, and copies of the examination files. Search terms and time ranges are documented. Tool versions, settings, errors, and output paths are recorded.

Analysis. Individual observations are correlated to test hypotheses. For example, a server log may show a successful login, an access record may show a file read, and a USB artifact may show that a device was connected near the same time. Investigators assess whether the combined

evidence supports unauthorized access and copying, while considering alternatives such as an authorized user, automated backup, incorrect clock settings, or shared credentials.

Presentation. The report states the question investigated, authority and scope, evidence received, methods, tools, integrity checks, findings, limitations, and conclusion. Technical terms are explained for non-technical decision-makers. Facts, interpretations, and unresolved questions are separated.



2. Investigation Flowchart

Complaint received and incident recorded

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Confirm authorization, scope, and preservation requirements

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Identify server, workstation, USB, accounts, logs, and relevant files

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Document scene and collect volatile evidence if appropriate

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Acquire forensic images / preserve logs and calculate hashes

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Verify integrity and maintain chain-of-custody records

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Examine copies: logs, file-system artifacts, USB artifacts, relevant files

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Correlate timestamps and test competing explanations

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Validate findings, peer review, and document limitations

↓

Prepare report and present findings to authorized college authority

3. Potential Evidence and Investigative Value

Evidence item	**Potential information**	**Investigative value**	**Caution**
Administrator's complaint / incident ticket	Reported discovery, affected files, approximate time window, authorized access expectations	Contextual / lead-generating	A report is an allegation, not proof of the event.
Server authentication logs	Account names, login outcomes, source addresses, session times	Potentially direct for recorded authentication events	Logs may be incomplete, altered, or affected by clock drift.
Server file-access or audit logs	File path, operation, account, time, success/failure	Potentially direct for logged file access	A read event does not by itself prove copying or identify the human operator.

Suspect workstation forensic image	User profiles, recent files, shortcuts, event logs, browser artifacts, USB history	Corroborating and potentially direct for local activity	Shared use and remote control are alternative explanations.
Recovered USB device and image	File copies, filesystem metadata, volume identifiers, deleted remnants	Potentially strong corroboration of transfer	Presence of a file on USB does not alone prove who copied it or when.
Examination-related files	Contents, names, hashes, metadata, access permissions	Identifies the data allegedly affected	Preserve confidentiality and minimize exposure.

4. Collection and Preservation Precautions

- Obtain appropriate written authorization and define the scope, including which systems, accounts, dates, and data may be examined.
- Photograph and document the scene, cable connections, device state, displayed time, serial numbers, and attached peripherals before changing anything.
- Do not browse files casually or use the suspect workstation for ordinary tasks. Any interaction can alter timestamps, logs, caches, and other artifacts.
- Consider volatile evidence such as active sessions, running processes, and network connections before shutdown, following the organization's approved response procedure.
- Use a validated hardware or software write blocker for storage acquisition where applicable. If live acquisition is necessary, document why it was selected and what changes it may cause.
- Create bit-for-bit forensic images when feasible, record acquisition errors and bad sectors, and calculate cryptographic hashes such as SHA-256 for verification.
- Seal original devices in suitable evidence packaging, label them clearly, control access, and store them in a secure evidence room.
- Maintain a contemporaneous chain-of-custody record with each transfer, date/time, purpose, sender, receiver, and signatures.

- Keep original evidence read-only and conduct analysis on verified working copies. Restrict access to confidential examination materials.
- Record tool name/version, settings, commands, examiner identity, system time, time zone, and any deviations from standard procedure.

```
PS C:\Users\angel> Get-FileHash C:\CDF_FOLDER\exam_paper.pdf -Algorithm SHA256

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          C008D4A3901A408B1F27E26F6B545CEF481D42A91647B0D13FC398A5273B0773       C:\CDF_FOLDER\exam_paper.pdf

PS C:\Users\angel>
```

5. Establishing a Timeline

A timeline combines events from independent sources into one normalized sequence. Investigators should preserve original timestamps and separately record normalized time, including the source time zone and any known clock offset. Relevant events may include account login, file open/read, USB insertion, file creation on the USB, logout, and later discovery.

```
PS C:\Users\angel> Get-ChildItem C:\CDF_FOLDER\ | Select Name,CreationTime,LastWriteTime,LastAccessTime

Name          CreationTime        LastWriteTime       LastAccessTime
----          ------------        -------------       --------------
exam_paper.pdf 29-10-2026 19:22:48 29-10-2026 19:22:49 29-10-2026 19:22:49


PS C:\Users\angel>
```

Illustrative time (local)	**Source / artifact**	**Observed event**	**Interpretation and confidence**
21:04:12	Server authentication log	Account A successfully authenticated	Supports a session beginning; does not identify the human user alone.
21:07:35	Server audit log	Account A accessed Exam_Paper.pdf	Direct evidence of a logged access if log integrity is established.

21:11:02	Workstation USB artifact	USB volume identifier recorded	Supports connection of a particular device around this time.
21:13:46	USB filesystem metadata	Exam_Paper.pdf exists on USB image	Supports possible transfer; timestamp semantics and prior presence must be assessed.
08:20:00 next day	Administrator complaint	Unexpected access reported	Establishes reporting time, not necessarily event time.

These times are hypothetical examples only. Investigators compare file hashes to determine whether the server and USB files are identical, examine copy-related artifacts, check clock synchronization and daylight-saving/time-zone effects, and seek corroboration from independent sources. A timeline should include uncertainty rather than presenting every timestamp as exact.

6. Final Forensic Report and Non-Technical Presentation

- Case identifier, investigator details, date of report, requesting authority, and authorization/scope.
- Executive summary: allegation, principal findings, and conclusion in plain language.
- Evidence inventory and chain-of-custody summary.
- Acquisition and preservation methods, write-protection details, hashes, and integrity verification.
- Tools and versions, examination steps, search criteria, and reproducibility details.
- Findings with artifact paths, timestamps, account identifiers, hash values, and supporting screenshots/tables where appropriate.
- Timeline and correlation of independent evidence sources.
- Alternative explanations, limitations, missing data, and confidence level.

- Conclusion that answers the investigation questions without overstating what the evidence proves.
- Appendices containing detailed logs, hash manifests, chain-of-custody forms, and relevant screenshots.

For non-technical authorities, use a one-page summary, a simple timeline, and a table that separates 'what was observed' from 'what it may mean.' Avoid jargon or define terms such as hash, forensic image, and metadata. State clearly whether a finding is confirmed, supported, inconclusive, or outside the examination scope.

7. Legal and Ethical Considerations

Investigators must act under valid institutional policy and applicable law, use only the access authorized, and avoid examining unrelated personal information. Confidential examination materials and student/staff records require strict access control and secure storage. Evidence should not be altered, planted, selectively reported, or shared with unauthorized persons. Investigators must document methods transparently, preserve exculpatory as well as inculpatory evidence, maintain impartiality, and disclose limitations. If evidence may be used in disciplinary or legal proceedings, collection and custody records must be sufficient to demonstrate authenticity and continuity.

Conclusion

A defensible investigation relies on lawful authority, careful preservation, verified forensic copies, independent evidence sources, and a transparent timeline. Server records may show access, while workstation and USB artifacts may corroborate possible copying. No single artifact should be treated as conclusive proof of a person’s intent without considering alternative explanations.

ASSIGNMENT 2(2)

Linking a Suspicious File to a User Using File-System Evidence

Case Background

A shared desktop computer contains a suspicious executable, hidden files, encrypted documents, browser artifacts, and several user accounts. Investigators want to determine whether the artifacts can be linked to a particular account and use a forensic tool such as ILookIX to examine the evidence.

8. Hardware and Software Inventory

Before examination, record the system's physical and logical configuration. This establishes what was examined and helps interpret artifacts correctly.

Category	Items to document	Why it matters
Computer hardware	Manufacturer, model, serial number, asset tag, CPU, RAM, BIOS/UEFI details	Identifies the system and may explain capability or acquisition constraints.
Storage	Drive type, make/model, serial number, capacity, interface, partition layout	Identifies the source media and its logical structure.
Peripherals	USB devices, external drives, card readers, docking stations	May provide transfer paths or relevant device identifiers.
Operating system	Edition, version/build, architecture, installation date, patch level	Affects artifact locations, logging, and timestamp interpretation.
File systems	NTFS, FAT/exFAT or other format; volumes and partitions	Determines metadata and recovery methods available.
Accounts and permissions	Local/domain accounts, SIDs, profile paths, group membership	Helps correlate artifacts with account profiles and access rights.

Applications	Browsers, security software, archive tools, encryption tools, remote-access tools	Explains possible artifact sources and alternate activity paths.
Forensic environment	Acquisition tool, ILookIX version, workstation, write blocker, hash algorithm	Supports repeatability and documentation.



9. How Storage, Operating Systems, File Systems and Metadata Help

Storage devices retain allocated files and may also contain deleted data, unallocated space, slack space, and remnants of previous files. The operating system records account profiles, application usage, logon activity, device connections, and security events. File systems such as NTFS maintain structures including directory entries, file records, timestamps, permissions, and identifiers. These structures can help locate a file and understand its history.

Metadata can include filename, full path, size, extension, creation/modification/access timestamps, owner or security identifier, permissions, file identifier, and cryptographic hash. Investigators compare these attributes with application artifacts such as shortcut files, recent-file lists, prefetch data, jump lists, browser downloads, antivirus logs, and event records. Metadata is useful but not infallible: timestamps can be changed, copied files may inherit or receive new times, and system behavior differs by file system and application.

10. Hidden and Encrypted Files

A hidden file is not necessarily malicious. It may be hidden through a file attribute, naming convention, application setting, or storage in a less obvious path. Investigators should enable forensic views of hidden/system files in the analysis tool, inspect directory metadata, search by file signature and known hashes, and examine unallocated space when justified. A file extension should not be trusted by itself; its signature and structure should be checked.

Encryption can prevent content review without the correct key or credentials. Investigators should preserve encrypted originals, record file metadata and hashes, identify the encryption software and relevant key stores where authorized, and avoid repeated guessing that could alter evidence or lock an account. They may seek keys from authorized credential stores, user-provided recovery material, enterprise key management, or memory acquisition where legally permitted. Any decryption attempt should be documented, and decrypted output should be treated as a derivative exhibit linked to the encrypted source.

11. Evidence-Linkage Table

Artifact	**Potential correlation**	**Link strength**	**Cautions**
Suspicious.exe in C:\Users\UserB\Downloads	Path is inside UserB profile; creation time aligns with browser download	Moderate when corroborated	A different person may have used the account or copied the file.
Browser download history	Shows URL, download time, target path, and possibly initiating profile	Strong for browser-recorded download; not necessarily human identity	History can be cleared, imported, or altered.
NTFS file record / metadata	Filename, timestamps, file identifier, owner/security descriptor	Supporting	Ownership and timestamps do not prove who physically acted.
Prefetch or application execution artifact, if present	May support execution of a program on the system	Corroborating	Availability depends on Windows version/configuration; not always proof of

			user intent.
Recent-file shortcut / Jump List	References the suspicious document and may include path and time-related data	Corroborating	Artifacts may be created by previews, indexing, or automated processes.
User logon/security events	Shows account session active near relevant times	Contextual; potentially strong for session attribution	A logged-in session may be unattended or remotely controlled.
USB device registry/artifact	Device identifier and connection history near file activity	Corroborating for device connection	Does not prove a specific file was copied.
Hash match between recovered file and known sample	Shows byte-level identity with the compared sample	Strong for file identity	Does not identify the person who obtained or executed it.

12. ILookIX Evidence Import and Analysis Workflow

The exact menus and capabilities may differ by ILookIX version, so the workflow below describes the intended forensic process rather than claiming that an examination was performed.

1. Confirm authority, case scope, evidence identifier, and chain-of-custody documentation.
2. Acquire or receive the forensic image and its hash manifest; preserve the original media and make a verified working copy.
3. Open the case in ILookIX and create a new case identifier. Enter examiner, date, evidence description, and notes.
4. Import the forensic image or supported evidence container using the tool's documented import/acquisition function. Select the correct source and avoid writing to the original media.

5. Verify the image integrity against the supplied hash, where supported. Record the algorithm, expected value, calculated value, and verification result.
6. Allow the tool to parse partitions, file systems, user profiles, deleted entries, metadata, and supported system/application artifacts.
7. Search for the suspicious filename, path fragments, known hash values, relevant extensions, and time window. Review hidden/system entries and deleted-file indicators.
8. Inspect account profiles and correlate the file with browser history, recent-file artifacts, execution traces, logon records, and removable-media artifacts.
9. Export only necessary artifacts as derivative copies. Record the original path, evidence ID, extraction method, timestamps, hash values, and export location.
10. Validate important findings with a second method or independent artifact, preserve screenshots/reports, and document parsing errors or unsupported formats.
11. Close the case with a clear list of findings, limitations, and unresolved questions. Keep the original evidence protected.

13. Direct Attribution vs Contextual Evidence

Evidence type	Example	Meaning
Direct evidence of an event	A valid audit record states that account SID X accessed a specific file at a recorded time.	Directly records the system event, subject to log reliability; does not automatically identify the human operator.
Strong corroboration	Browser download record and matching file in the same user profile at a consistent time.	Independent artifacts support the same account/activity interpretation.
Contextual evidence	A user's profile contains a file or the user was logged in during the period.	Shows opportunity or association, but not necessarily the action.

14. Limitations and Alternative Explanations

- The computer is shared; another person may have used the logged-in account.
- Credentials may have been shared, stolen, or used through remote access.
- Timestamps may reflect copying, restoration, clock drift, time-zone conversion, or application-specific behavior.
- Artifacts may be missing because of log rotation, cleanup, anti-forensic activity, system reinstallation, or unsupported file formats.
- A suspicious executable may have been downloaded automatically, quarantined, or placed by legitimate administrative software.
- File ownership or profile location does not prove authorship or intent.
- Encrypted content may not be available for review, limiting conclusions about its purpose.
- The forensic tool may parse some artifacts incorrectly or not support a particular Windows/application version; validation is required.

Conclusion

Attribution should be based on a combination of account-specific paths, reliable timestamps, file-system metadata, browser or execution artifacts, and session records. The investigator should distinguish evidence that directly records a system event from evidence that merely places a file in a user profile. A fair conclusion must acknowledge shared use, remote access, timestamp uncertainty, and tool limitations.

ASSIGNMENT 2(3)

Forensic Imaging and Safekeeping of a Suspect Storage Device

Case Background

A laptop seized during a suspected data-theft investigation contains an internal storage device and an attached USB drive. Investigators must collect and preserve both, acquire forensic images, verify integrity, consider data carving, and maintain safekeeping. IXImager and the ASB container format are identified for the proposed workflow.

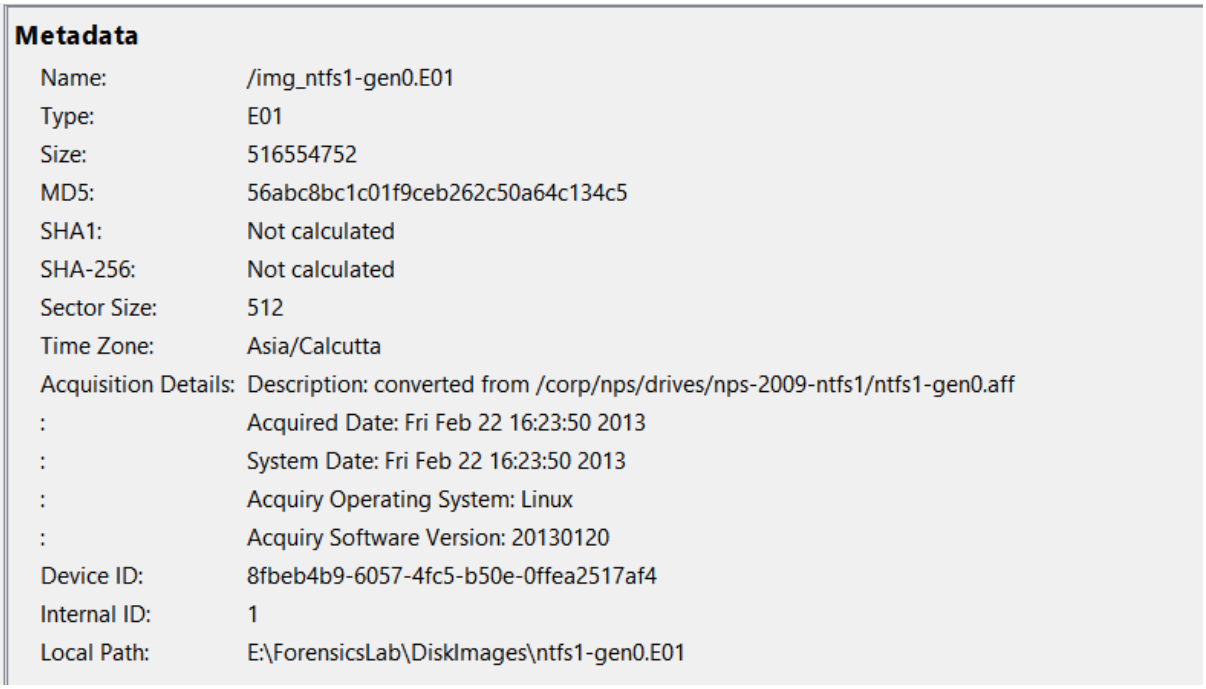
15. Safe Collection and Transport

12. Confirm legal/institutional authority and record the seizure location, date, time, persons present, and reason for seizure.
13. Photograph the laptop, screen state, cables, connected USB drive, ports, and any visible identifiers before disconnecting anything.
14. Record make, model, serial number, asset tag, power state, visible clock, network state, and attached peripherals.
15. Assess whether the system is powered on and whether volatile data is likely to be important. Follow the approved live-response procedure; do not casually power off or interact with it.
16. If powered off, do not boot it. If powered on, an appropriately trained examiner decides whether to collect volatile data before isolation or shutdown, documenting every action and risk.
17. Label the laptop and USB drive separately with unique evidence IDs. Record the USB's physical connection and port where practical.
18. Use suitable anti-static packaging for storage devices and protective packaging for the laptop. Keep devices secure and protected from impact, moisture, heat, and electromagnetic hazards where relevant.
19. Transport in a sealed, tamper-evident package. Record every transfer in the chain-of-custody form.
20. Store the items in a controlled-access evidence area until acquisition.

16. Physical Acquisition and Analysis of Images

Physical acquisition aims to obtain a sector-level copy of the source storage, including allocated files, file-system structures, unallocated space, and potentially recoverable remnants. A logical acquisition generally collects selected files or accessible file-system objects and may omit deleted data or unallocated sectors. The correct acquisition type depends on the case scope, device condition, encryption, and available tools.

Analysis should normally be performed on a forensic image because ordinary access to original media can change data, update system metadata, trigger background processes, or accidentally delete evidence. A verified image creates a repeatable working copy, allows multiple analysts to examine the same evidence, and protects the source from routine analysis operations. If bad sectors or acquisition errors occur, they must be recorded rather than concealed.



Metadata

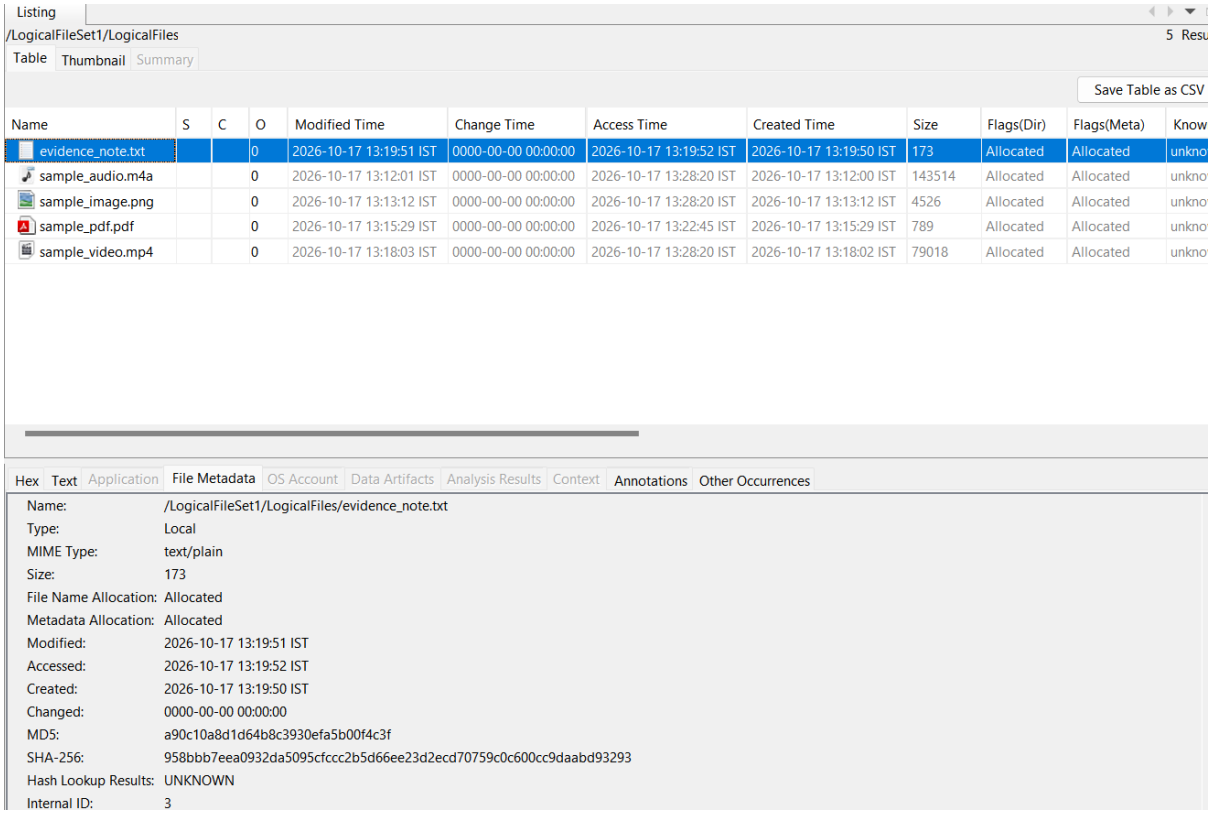
Name:	/img_ntfs1-gen0.E01
Type:	E01
Size:	516554752
MD5:	56abc8bc1c01f9ceb262c50a64c134c5
SHA1:	Not calculated
SHA-256:	Not calculated
Sector Size:	512
Time Zone:	Asia/Calcutta
Acquisition Details:	Description: converted from /corp/nps/drives/nps-2009-ntfs1/ntfs1-gen0.aff
:	Acquired Date: Fri Feb 22 16:23:50 2013
:	System Date: Fri Feb 22 16:23:50 2013
:	Acquiry Operating System: Linux
:	Acquiry Software Version: 20130120
Device ID:	8fbeb4b9-6057-4fc5-b50e-0ffea2517af4
Internal ID:	1
Local Path:	E:\ForensicsLab\DiskImages\ntfs1-gen0.E01

17. IXImager Forensic Imaging Workflow

The following is a proposed workflow aligned with the assignment prompt. Exact screens, supported formats, and verification options must be checked against the installed IXImager version and laboratory procedure.

21. Create a case record and assign unique evidence IDs to the laptop’s internal drive and USB device.

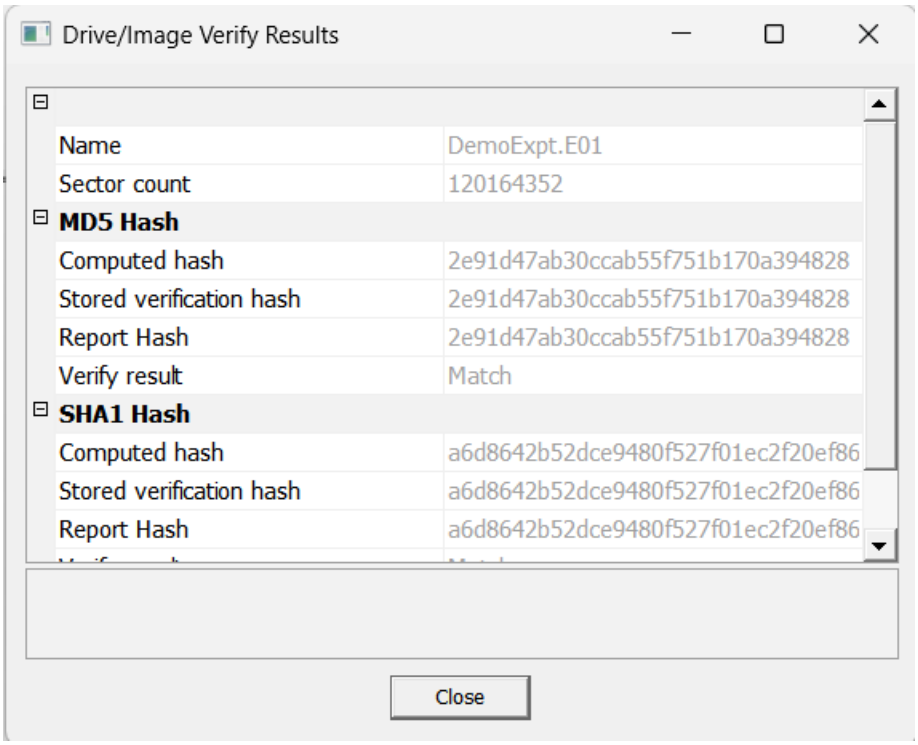
22. Document source make/model, serial number, capacity, interface, condition, and any encryption or hardware issues.
23. Connect the source through a validated write blocker where supported. Confirm the source is not mounted for ordinary read/write use.
24. Launch IXImager on the forensic workstation and select the correct physical source. Double-check the device identity and capacity before starting.
25. Choose the required acquisition type and output destination. Ensure the destination has sufficient free space and is not the source device.
26. Select the documented hash algorithm(s), such as SHA-256, and enable verification options available in the tool.
27. Start acquisition and record start/end times, operator, tool version, settings, output path, progress, errors, unreadable sectors, and any interruptions.
28. After acquisition, compare the tool's verification result and source/image hashes as supported. Record the exact hash values in the case notes.
29. Preserve the image as a master copy with restricted access. Create a separate working copy for examination and verify the working copy before use.
30. Generate an acquisition log and hash manifest. Seal, label, and return the original media to secure storage.
31. If using an ASB container, document how the container was created, what it contains, associated metadata, and how integrity was checked.



18. Hash Values and Integrity

A cryptographic hash is a fixed-length value calculated from data. If the data changes, the hash will normally change. A hash is therefore useful for demonstrating that an acquired image or exported file remains byte-for-byte consistent with the data used for comparison. SHA-256 is commonly used for integrity verification.

Before acquisition, a source hash may be calculated when the acquisition method and device condition permit it. During or after acquisition, the tool may calculate a hash for the image and/or verify the image against the source. Investigators record the algorithm and full values, not just a statement that the hashes matched. For damaged media, errors and unreadable sectors can affect whether a complete comparison is possible; this must be documented.



Stage	**Example record (illustrative only)**	**Interpretation**
Source identification	Internal SSD, evidence ID L-01, 512 GB	Identifies the intended source.
Acquisition hash	SHA-256: [record actual calculated value]	Must be filled from the tool output; do not invent a value.
Image verification	Verification result: Pass / Fail / Partial (as actually reported)	A pass supports integrity for the verified data; partial results require explanation.
Working copy verification	SHA-256 of master and working copy recorded	Matching values support that the working copy matches the master image.

19. ASB Container and Accompanying Information

The assignment identifies the ASB container format as part of the evidence workflow. Because container behavior can depend on the specific forensic product and version, the examiner should confirm the format's exact purpose and supported metadata in the applicable IXImager/ASB documentation rather than assuming features that have not been verified. In a forensic workflow, an evidence container may package an image or related evidence with case metadata and integrity information.

The evidence record accompanying a container should include the case ID, evidence ID, source device details, acquisition date/time and time zone, examiner, acquisition tool/version, acquisition type, hash algorithm and values, verification outcome, any read errors, container filename and size, encryption/password handling if applicable, chain-of-custody entries, and a description of included items. The container itself should be stored securely and checked for integrity after transfer.

20. Data Carving

Data carving is the recovery of files or file fragments by searching raw data for recognizable file signatures, structures, or patterns without relying entirely on directory entries. It is useful when file-system metadata is damaged, deleted, overwritten, or unavailable. For example, an analyst may search unallocated space for known document, image, or archive signatures.

Carving can recover content without its original filename, folder path, owner, or reliable timestamps. Fragmented files may be incomplete or reconstructed incorrectly, and signature collisions can produce false positives. Each carved result should be recorded with its source offset, method, tool/version, recovery status, and hash. The analyst should validate file structure and compare with other evidence before interpreting it.

21. Date/Time Problems

- Record the device clock, laboratory clock, time zone, UTC offset, and daylight-saving status where applicable.
- Distinguish timestamps stored in UTC from those displayed in local time; document any conversion.
- Check for clock drift by comparing device time with reliable external events or server logs, while accounting for network time synchronization.

- Understand timestamp semantics: creation, modification, access, metadata-change, download, and log-event times are not interchangeable.
- Consider copying, extraction, restoration, backup, and file-system operations that may alter or preserve timestamps differently.
- Preserve raw timestamp values and show normalized values separately in the timeline.
- Express uncertainty explicitly if the source time zone or clock accuracy cannot be established.

Conclusion

Reliable imaging depends on correct source identification, write protection, documented acquisition, hash verification, secure storage, and a complete chain of custody. Data carving may recover material that file-system analysis misses, but carved content often has weaker contextual information and must be validated carefully.

ASSIGNMENT 2(4)

Recovery of Deleted Evidence from Volume Shadow Copies

Case Background

An employee is suspected of deleting documents and browser-related artifacts after unauthorized activity. A forensic image of a Windows system is available. Volume Shadow Copies are present, and investigators need to select relevant evidence, analyze events, compare earlier versions, validate recovered artifacts, and reconstruct the sequence of activity.

23. Selecting and Categorizing Evidence

Investigators begin by defining the incident question, relevant users, date range, file types, applications, and suspected activity. They inventory the image and identify partitions, user profiles, relevant Windows logs, browser databases, documents, deleted-file entries, Volume Shadow Copy snapshots, and security artifacts. Evidence should be prioritized according to relevance, reliability, uniqueness, and ability to answer the investigative question.

A triage plan prevents unnecessary review of unrelated personal data. Analysts preserve original artifacts and record the path, evidence ID, source snapshot, hash, timestamp interpretation, and reason for selection. Potentially exculpatory evidence should not be discarded. Irrelevant data should be excluded from detailed review only after the exclusion basis is documented.

24. Evidence Classification Scheme

Category	Examples	Purpose / treatment
User-created files	Documents, spreadsheets, scripts, downloaded files	Compare versions, metadata, hashes, and paths; protect confidential content.
System/application files	Windows event logs, registry hives, browser databases, application configuration	Provide system and application context; preserve structure and metadata.
Potentially suspicious files	Unknown executable, renamed archive, unusual	Prioritize for signature/hash analysis and behavior/context

	script, files in temporary paths	review; avoid executing on a live system.
Deleted/recovered artifacts	Deleted directory entries, recovered documents, browser database remnants, carved files	Record recovery method and limitations; validate content and provenance.
Supporting/contextual evidence	Logon records, shortcut files, USB artifacts, backup records, network logs	Corroborate the sequence and test alternative explanations.

25. Event Analysis

Event Analysis organizes artifacts as discrete events and places them in a chronological sequence. Events may come from Windows Security logs, application logs, file-system metadata, browser history, removable-device records, and snapshot comparisons. A useful event record contains the source, event identifier or artifact path, raw timestamp, normalized timestamp, account, action, object, and confidence.

Analysts compare events from independent sources to see whether they support a consistent explanation. For example, a file modification event followed by a USB connection and a deletion event may suggest a sequence, but the interpretation must be checked against application behavior, clock differences, and whether the events record an actual user action or an automated process.

26. Cloud Analysis and Lead Analysis

Cloud Analysis examines authorized cloud-related artifacts and records, such as synchronization logs, cloud drive metadata, account activity, browser access, file-sharing records, and provider audit logs. It may help answer whether a file was uploaded, synchronized, shared, downloaded, or accessed from another device. Investigators must respect authorization, retention limits, provider time zones, and the difference between a local sync event and a confirmed server-side action.

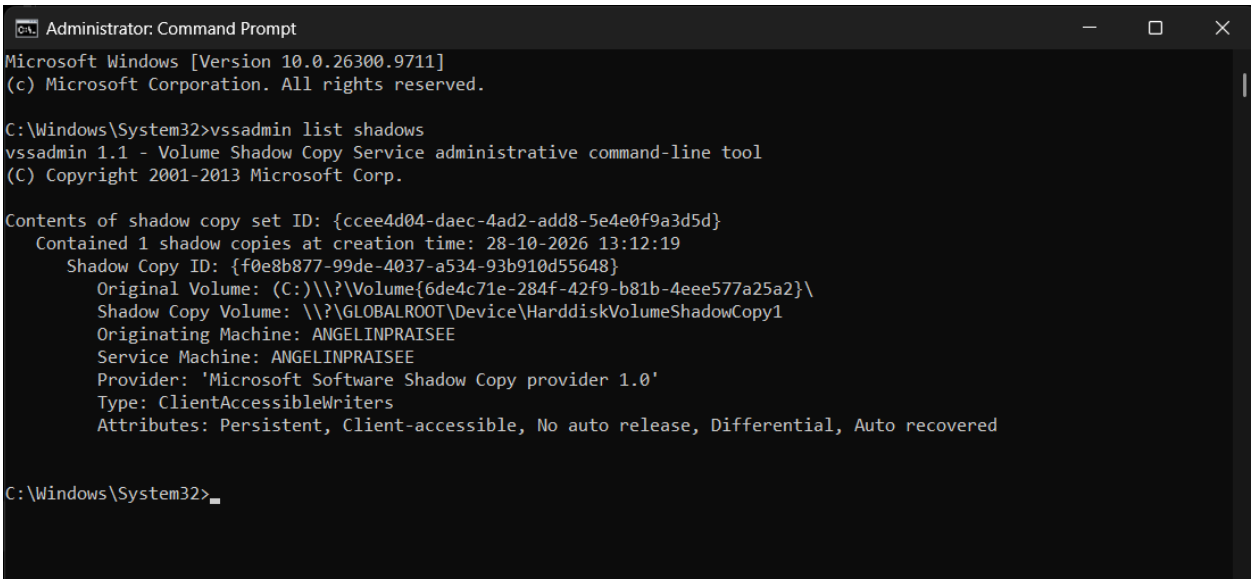
Lead Analysis uses an initial artifact to generate a testable investigative direction. A suspicious filename may lead to a search for matching hashes, related paths, browser downloads, email attachments, USB records, or a previous snapshot. A lead is not itself a conclusion. Each lead should be recorded with its origin, proposed question, follow-up procedure, result, and whether it was corroborated or rejected.

27. Volume Shadow Copy Analysis

Volume Shadow Copy Service (VSS) can maintain point-in-time snapshots of a Windows volume for functions such as restore points and previous versions. If a relevant file existed when a snapshot was created, a copy or earlier version may remain available even after the live file is modified or deleted. Investigators should enumerate available snapshots from the forensic image using a suitable forensic tool, preserve snapshot metadata, and examine the relevant prior paths without altering the image.

32. Identify Windows volumes and enumerate available shadow-copy snapshots and their creation times.
33. Record snapshot identifiers, source volume, timestamps, and tool used to access them.
34. Locate the target file or artifact in the live file system and in relevant snapshots.
35. Export the earlier version from the forensic working copy, preserving its source snapshot and path.
36. Calculate a hash for each extracted version and compare contents, metadata, and relevant timestamps.
37. Examine whether a browser database, log, or other artifact also has an earlier version in a snapshot.
38. Document missing snapshots, parsing failures, and any uncertainty about snapshot completeness.

A snapshot is not guaranteed to contain every deleted item. Snapshots may have been removed, overwritten, or created after the event; some application data may not have been captured in a useful state. Recovery should therefore be described as a result of the available snapshot set, not as proof that all historical activity has been reconstructed.

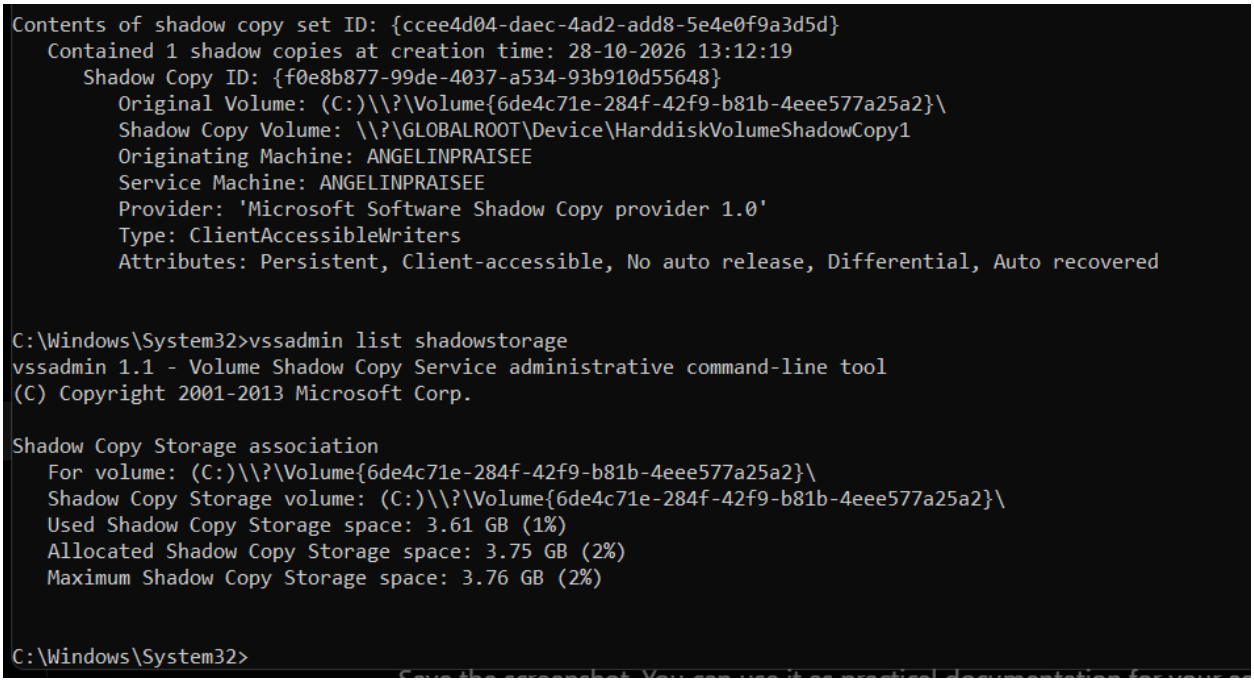


28. Hypothetical Timeline

Time (illustrative)	Artifact/source	Event	Interpretation
09:12	File-system metadata	Report.docx first observed in user profile	Potential creation/arrival time; verify timestamp semantics.
09:18	Office/application artifact	Document opened or edited	Supports interaction if the artifact reliably records it.
09:31	USB connection record	Removable drive connected	Shows device connection, not necessarily a copy.
09:34	USB image	Matching document hash found on USB	Supports presence of an identical file on the device.
09:47	Live file system	Report.docx no longer present at original path	Could reflect deletion, move, rename, or cleanup.

Snapshot created earlier at 09:00	Volume Shadow Copy	Earlier version of Report.docx recovered	Shows the file existed in that snapshot; it does not by itself prove who deleted it.
10:05	Windows/browser logs	Additional activity observed	Must be correlated with independent evidence and clock accuracy.

This timeline is hypothetical and illustrates the method only. A real timeline should use actual extracted artifacts, include raw and normalized timestamps, identify the time zone, and state confidence and uncertainty for each event.



29. Validating Recovered Artifacts

- Confirm the source image and snapshot were acquired and verified correctly.
- Record the exact snapshot ID, source volume, original path, extraction tool, version, and extraction time.

- Calculate a cryptographic hash of the recovered artifact and preserve the exported file as a derivative exhibit.
- Open or parse the recovered item using a suitable viewer without modifying the preserved copy; verify that its structure and content are plausible.
- Compare it with other versions using hashes, binary comparison, metadata, or content differences.
- Corroborate with independent artifacts such as event logs, browser records, application history, or matching files on other devices.
- Check timestamp interpretation, time zone, file-system behavior, and possible restoration or synchronization effects.
- Document limitations, recovery errors, incomplete fragments, and any assumptions.

30. Sources of Error and Alternative Explanations

- Clock drift, incorrect time zone, daylight-saving conversion, or inaccurate system clock.
- Different timestamp semantics, including creation, modification, metadata-change, and access times.
- File copying, backup, restore, synchronization, or extraction changing or preserving timestamps in unexpected ways.
- A file may have been moved or renamed rather than deleted.
- A recovered file may come from a different path, user, or earlier version than assumed.
- VSS snapshots may be incomplete, overwritten, disabled, or created at irregular intervals.
- Deleted-file recovery or carving may return partial data, fragments, or false positives.
- Automated software, antivirus, indexing, or synchronization may create artifacts without direct user action.
- A shared or unattended account complicates attribution to a particular person.

Conclusion

VSS analysis can restore historical context by allowing investigators to compare earlier versions with the live image. Strong findings require a traceable recovery method, hash validation, event correlation, and careful treatment of timestamp uncertainty.

ASSIGNMENT 2(5)

Multi-Source Investigation of a Suspected Cyber Incident

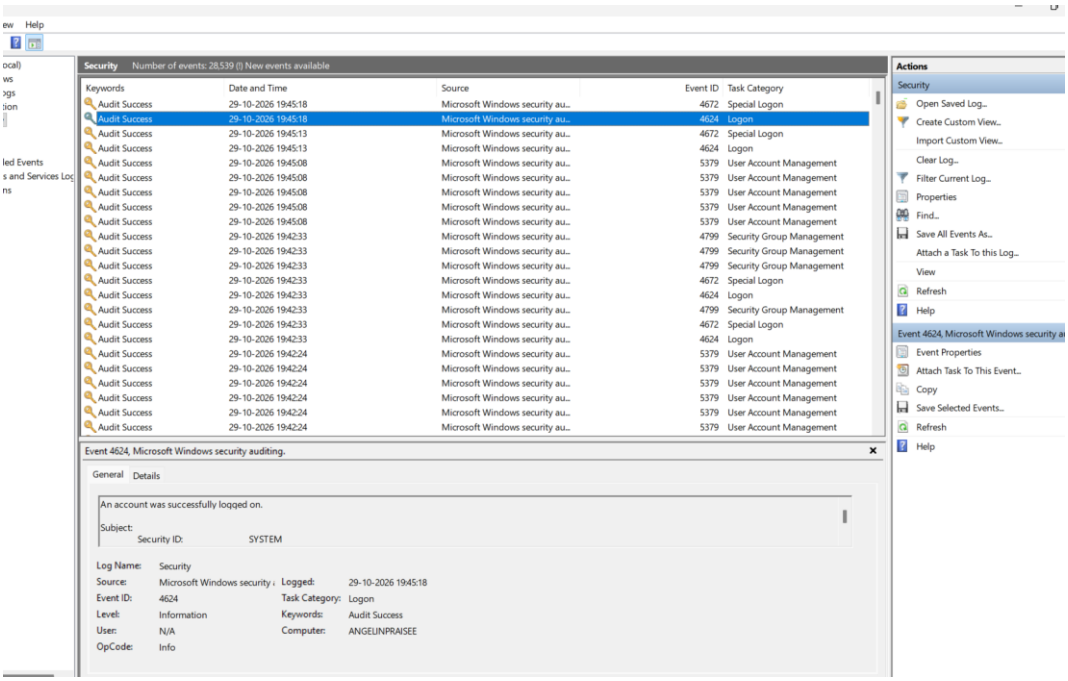
Case Background

A student account is suspected of being used in unauthorized access to an online service. With proper authorization, investigators can examine a Windows computer, browser artifacts, email and messaging records, a mobile phone, and relevant Internet/cloud records. They must correlate evidence across sources and account for the challenges of modern digital forensics.

31. Potential Evidence from Windows and Other Operating Systems

On Windows, potentially relevant sources include Security and System event logs, application logs, registry hives, user profile folders, NTFS metadata, browser databases, shortcut files, Jump Lists, prefetch artifacts where available, scheduled tasks, installed applications, USB connection artifacts, firewall records, and security-tool alerts. The exact artifacts depend on the Windows version, logging configuration, retention, and user activity.

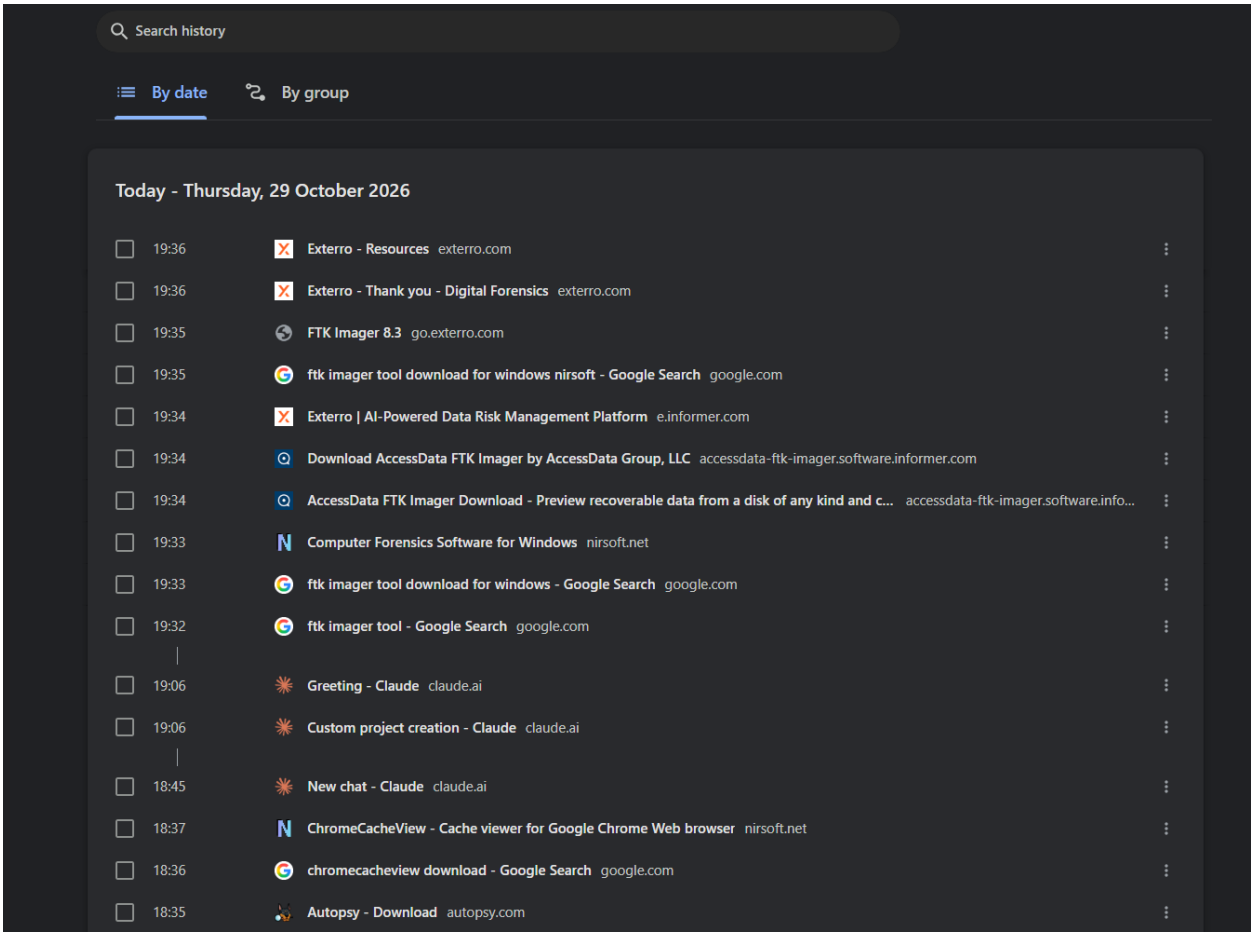
Other operating systems may contain analogous sources: macOS unified logs, user Library data, browser databases, file metadata, login records, and removable-device traces; Linux authentication logs, shell history, system journal, application logs, browser profiles, and file-system metadata. Shell history or application artifacts can be incomplete or altered and should not be treated as a complete record of activity.



32. Browser Artifacts and Cautions

Browser artifacts may include history, downloads, cache, cookies, session data, saved form data, bookmarks, extensions, autofill records, and timestamps. They can help identify websites visited, files downloaded, accounts used, or the sequence of browsing activity. Browser databases may also contain URLs and records associated with web-based applications.

Browser evidence must be interpreted carefully. A URL in history does not prove that the user completed an action, and a cached item may have been loaded automatically. Private browsing, cleanup, synchronization, profile sharing, extensions, malware, and multiple users can affect the artifacts. Investigators should correlate browser records with server-side logs, downloaded files, authentication records, and other independent sources.



33. Email and Messaging Evidence

Email evidence may include sender and recipient addresses, subject, body, message IDs, timestamps, routing headers, attachment names and hashes, mailbox folders, deleted messages, and server audit records. Messaging evidence may include conversation content, participants, message IDs, delivery/read states, timestamps, media attachments, backups, and application databases, depending on the platform and authorization.

Correlation can be performed by matching attachment hashes to files on the computer or mobile phone, comparing message timestamps with browser sessions and logins, and matching account identifiers across services. Email headers may help assess routing and authenticity. Investigators must account for forwarded messages, time-zone differences, unsent drafts, deleted or edited messages, and end-to-end encryption. A screenshot alone may omit metadata or be manipulated, so native records and provider data should be considered where lawfully available.

34. Mobile Phone Forensics and Challenges

Mobile examination begins with legal authorization, device identification, documentation of power/network state, and a decision about isolation and acquisition. Investigators record make/model, serial or IMEI where appropriate, operating-system version, SIM/eSIM details when relevant, storage capacity, encryption status, and visible condition. They should follow validated procedures and avoid unnecessary interaction that could change application state or trigger remote wipe.

Depending on the device and authorization, acquisition may be logical, file-system based, or physical. Relevant sources may include calls, SMS/MMS, messaging databases, contacts, photographs, application data, browser history, location records, notifications, and backups. Each acquisition type has limits; modern encryption and secure hardware may prevent full physical access.

Challenges include device lock codes, encryption, changing application formats, cloud-only content, app sandboxing, deleted or ephemeral messages, remote wipe, network synchronization, fragmented backups, limited battery, proprietary connectors, and the risk of altering evidence during handling. The examiner should preserve the device state where feasible, document every interaction, maintain chain of custody, and distinguish device-derived evidence from provider-derived records.

35. Internet and Cloud-Based Evidence

- Service-provider authentication and audit logs showing account access, IP address, session, and security events.
- Cloud storage metadata, file version history, sharing permissions, upload/download records, and synchronization logs.
- Email provider records, message headers, mailbox audit logs, and attachment information.
- DNS, proxy, firewall, VPN, and network connection logs where retained and lawfully accessible.
- IP address allocation records and relevant time-stamped records that may help interpret an observed address.
- Online-service logs showing the actions performed on the affected account, such as login, password reset, file access, or settings change.

- Backups, account recovery records, and security notifications that may corroborate access or account changes.

Cloud evidence presents jurisdiction, retention, provider cooperation, time-zone, and account-attribution challenges. An IP address generally identifies a network endpoint or subscriber connection, not necessarily the individual who performed an action. Investigators must distinguish a local synchronization event from a confirmed cloud-side operation and preserve provider exports with their provenance and integrity information.

36. Cross-Source Correlation Table

ID	Source	Hypothetical artifact	Correlation	Interpretation / limitation
A1	Windows	User profile contains downloaded file incident_notes.pdf	Hash matches email attachment A3	Supports file identity across sources; does not prove who opened it.
A2	Browser	History shows access to the affected online service at 14:02	Matches provider login record A6 within clock uncertainty	Corroborates access to the service; browser history alone does not prove a completed transaction.
A3	Email	Message contains incident_notes.pdf attachment sent at 13:55	Attachment hash matches A1	Supports that the same file was attached; verify message authenticity and account ownership.

A4	Messaging / mobile	Message references a service account and is timestamped 14:05	Temporal relationship to A2 and A6	Contextual unless message provenance and participant attribution are validated.
A5	Mobile	Authentication notification received at 14:03	Matches provider security event A6	Supports that a notification was delivered to the device; not proof of who initiated the login.
A6	Cloud/provider	Audit record shows account login at 14:02 from an IP address	Correlates with A2 and A5	Potentially strong evidence of a recorded login; IP does not alone identify the human actor.
A7	Windows security log	Account session active from 13:50 to 14:20	Overlaps A2 and A3	Provides session context; another person or remote actor may have used the session.

All records in the table are hypothetical examples, not findings from an actual device. Correlation should use exact artifact identifiers, raw timestamps, time zones, hashes, account IDs, and confidence ratings.

37. Major Challenges in Multi-Source Digital Forensics

Challenge	**Effect on investigation**	**Mitigation**

Different time zones and clock drift	Events may appear out of sequence.	Preserve raw times, record offsets, normalize carefully, and state uncertainty.
Encryption and locked devices	Content may be inaccessible.	Use lawful acquisition options, preserve keys/credentials when authorized, and document limitations.
Cloud provider control and jurisdiction	Evidence may be remote, transient, or legally restricted.	Use proper legal process, preserve provider identifiers, and request relevant audit records promptly.
Data volume and heterogeneity	Large amounts of data from incompatible formats slow analysis.	Use triage, standardized evidence IDs, validated parsers, and focused search criteria.
Ephemeral and deleted data	Messages, sessions, or logs may disappear quickly.	Prioritize preservation, collect volatile evidence appropriately, and document gaps.
Attribution and shared accounts	Account or device activity may not identify the person.	Correlate independent evidence and test alternatives such as shared use or remote access.
Tool limitations and parser errors	Artifacts may be missed or misinterpreted.	Record versions, validate important results independently, and retain raw exports.
Privacy and scope constraints	Unrelated personal data may	Minimize collection and

	be exposed.	review, restrict access, and follow authorization limits.
Integrity and chain of custody	Evidence may be challenged if handling is undocumented.	Hash evidence, maintain custody logs, secure originals, and document each transfer.
Different retention policies	Logs may cover different periods or omit events.	Map source coverage and clearly state which periods and sources were unavailable.

38. Final Investigative Conclusion

The final conclusion should distinguish the strength and role of each evidence category rather than treating all artifacts equally.

Category	**Illustrative evidence**	**How to report it**
Direct evidence	A validated provider audit record recording a successful login or a server-side action on the affected account.	State exactly what the record shows, its provenance, and its limitations. Avoid equating an account event automatically with a named individual.
Corroborating evidence	Browser history at a consistent time, a matching downloaded file, a mobile security notification, and a related message.	Explain how independent sources align and whether their timestamps and identifiers are reliable.
Contextual evidence	A user was logged into Windows, a device was connected to a network, or a message discussed the incident.	Use as context or a lead; do not present it as proof of the central action by itself.

Unresolved questions	Who physically operated the device, whether credentials were compromised, whether an IP was shared, or whether a log gap hides other activity.	List additional evidence needed and state that no conclusion is reached where support is insufficient.

A defensible final summary may state that the available artifacts are consistent with a particular sequence of account and device activity, while identifying which events are directly recorded and which are inferred through correlation. It should not name a person as responsible unless the evidence supports that attribution and reasonable alternatives have been assessed. Where records are missing or ambiguous, the report should say so plainly.

Conclusion

Multi-source investigations are strongest when artifacts from computers, browsers, communications, mobile devices, and cloud providers independently support the same sequence. Investigators must preserve each source, align timestamps carefully, validate extracted records, respect privacy and legal authority, and distinguish recorded account activity from proof of the individual behind it.